Advanced machine learning

Active learning

Transductive active learning: ITL and safe Bayesian optimization

Joachim Buhmann and Carlos Cotrini

Agenda

Model validation and uncertainty quantification

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- ML models
- Statistical estimators
- Linear estimators
- Neural networks
- Transductive active learning
- Batch active learning

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Model validation and uncertainty quantification

ML models	Cross-validation ✓
Statistical estimators	Bootstrap ✓
Linear estimators	Wald test ✓
Neural networks	Bayesian neural networks ✓
Transductive active learning	ITL and SafeBO
Batch active learning	ProbCover

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  - Diagnoses of rare diseases.
  - Autonomous driving.
  - Corporate fraud and data leakage.
- How can we collect a small yet representative sample when we are under a strict budget?

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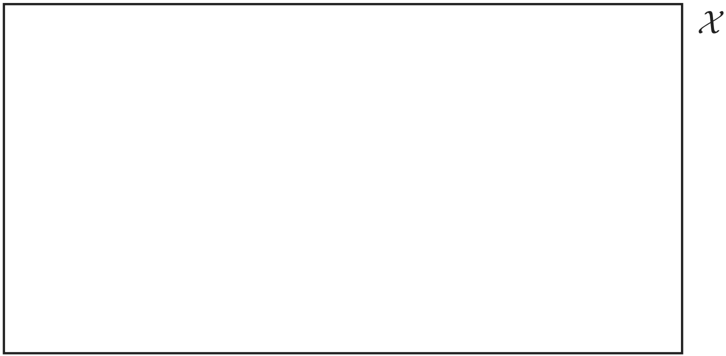
- How much information does a point (x_n, y_n) provide about f?
- How do we select $x_n \in \mathcal{S}$ to gain the most about f in $\mathcal{A}$?

The goal, geometrically



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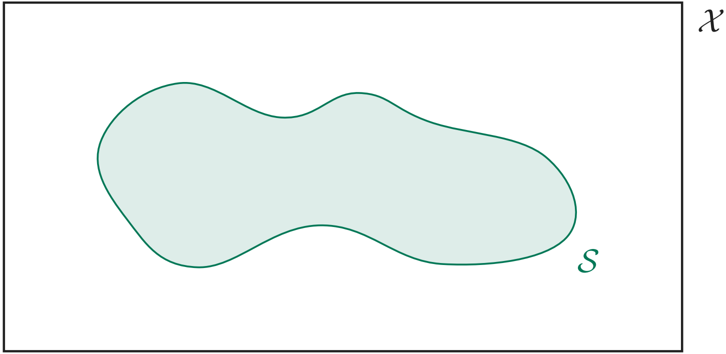

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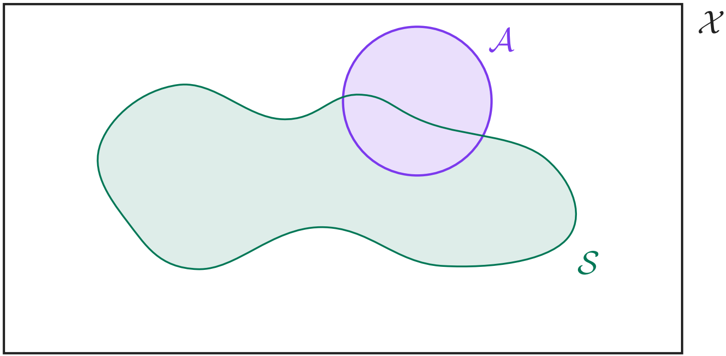

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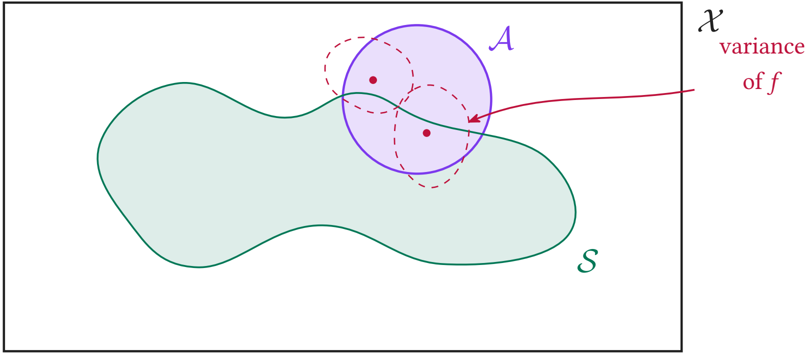

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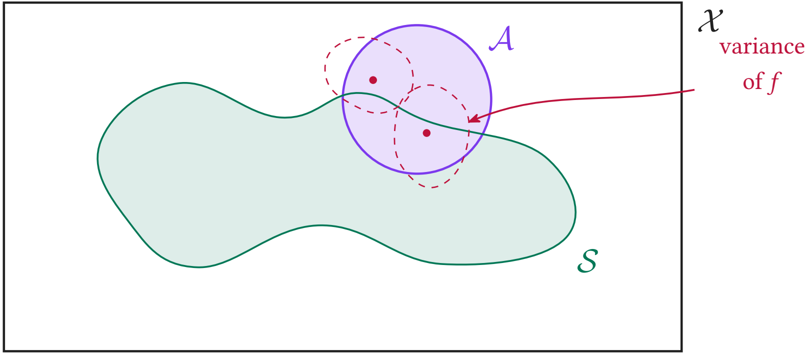

Goal: find $x_n \in \mathcal{S}$ most informative about f on $\mathcal{A}$.

Information-based transductive learning

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We pick $x_n \in \mathcal{S}$ to maximize the conditional mutual information between the observation $y_x = f_x + \epsilon_x$ and f restricted to the target $\mathcal{A}$.

$$x_n = \operatorname*{arg\,max}_{x \in \mathcal{S}} \; I(f_{\mathcal{A}} \,;\, y_x \mid \mathcal{D}_{n-1})$$

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$$I(f_{\mathcal{A}} \, ; \, y_x \mid \mathcal{D}_{n-1}) = \tfrac{1}{2} \log\left(\frac{\mathrm{Var}(y_x \mid \mathcal{D}_{n-1})}{\mathrm{Var}(y_x \mid f_{\mathcal{A}}, \mathcal{D}_{n-1})} \right)$$

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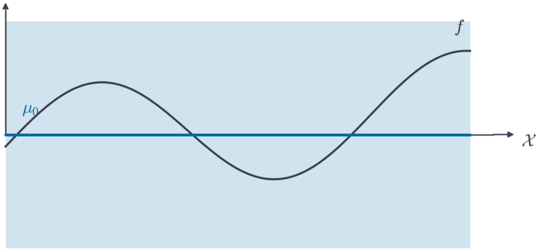
The identity holds for $f \sim \mathcal{GP}(\mu, k)$ with known μ, k (numerator: uncertainty in y_x before knowing f on $\mathcal{A}$; denominator: after).

Method: fit and update a GP

- Impose $f \sim \mathcal{GP}(\mu, k)$ on $\mathcal{X}$, with $\mu\colon \mathcal{X} \to \mathbb{R}$ and $k\colon \mathcal{X} \times \mathcal{X} \to \mathbb{R}$.



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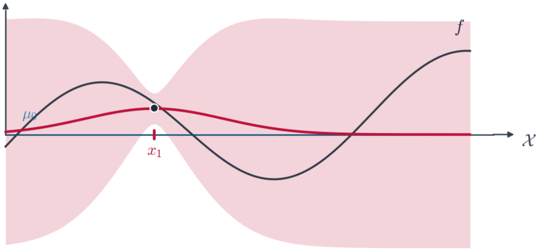

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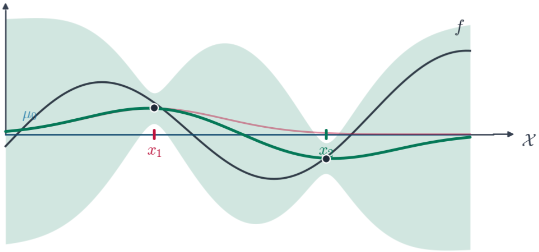
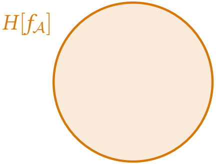
- For $i = 1, 2, \ldots$: pick $x_i \in \mathcal{S}$, observe y_{x_i}, update the posterior with (x_i, y_i).



The information-theoretic picture



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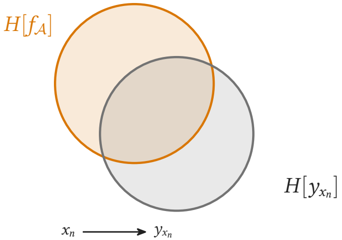

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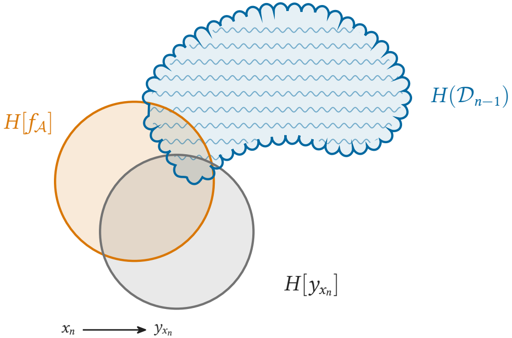

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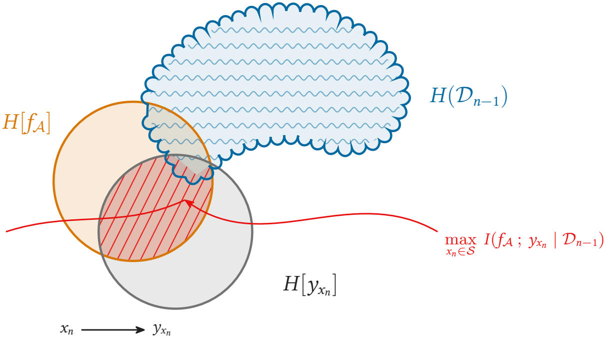

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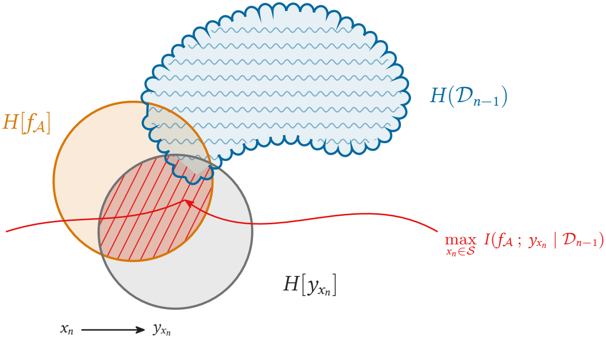

When $f \sim \mathcal{GP}(\mu, k)$:

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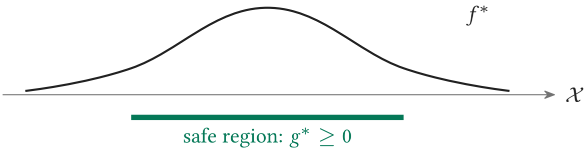

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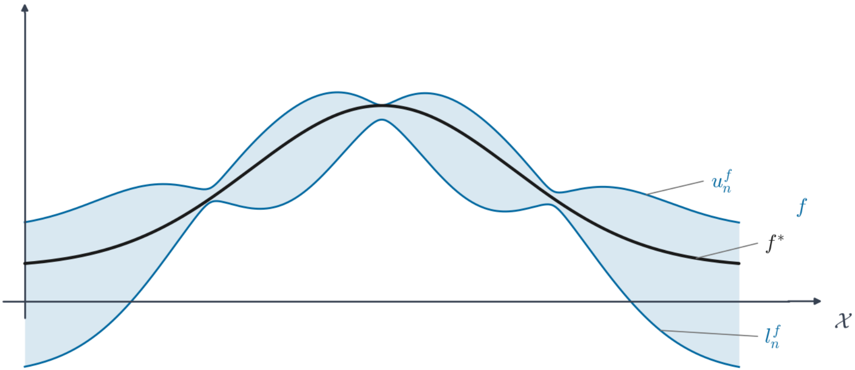

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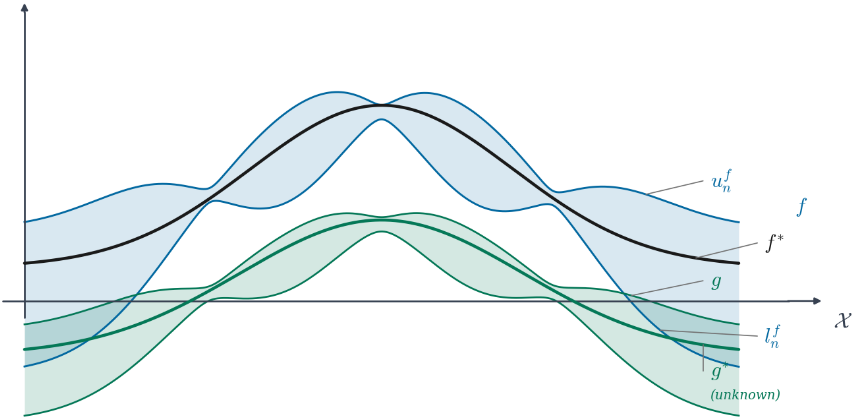

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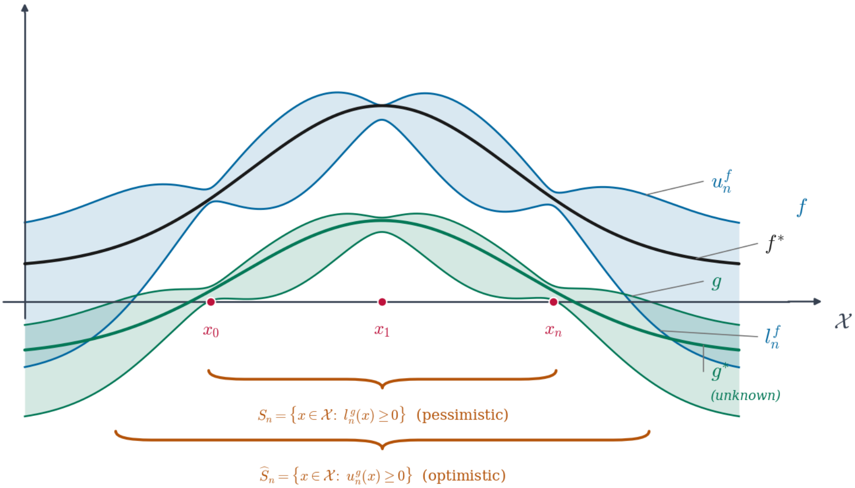

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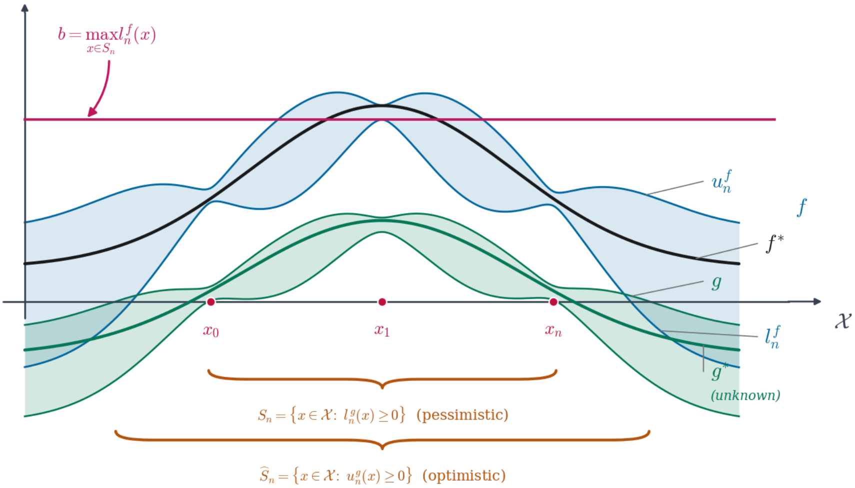

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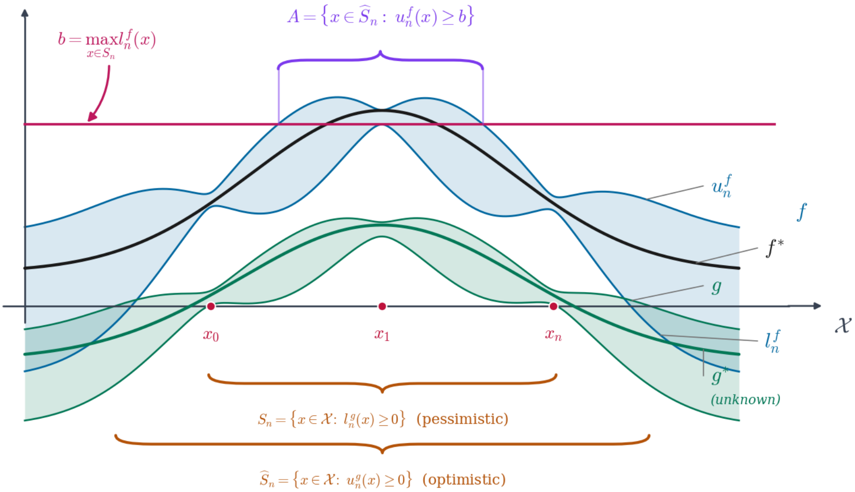

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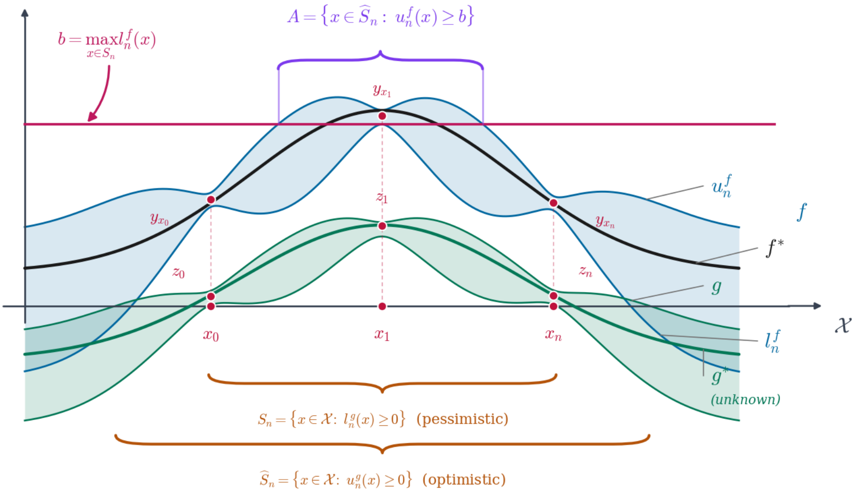

Safe Bayesian optimization with ITL

- Fit a GP f on $\{(x_i, y_i)\}_{i<n}$: it induces l_n^f, u_n^f with $[l_n^f(x), u_n^f(x)]$ a 95% confidence interval of $\mathbb{E}[f(x)]$.

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- Fit a GP g on $\{(x_i, z_i)\}_{i<n}$: it induces l_n^g, u_n^g, hence safe-set estimates

$$\mathcal{S}_n = \{x : l_n^g(x) \geq 0\} \text{ (pessimistic)}, \qquad \widehat{\mathcal{S}_n} = \{x : u_n^g(x) \geq 0\} \text{ (optimistic)}.$$

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- Then do ITL using sample space $\mathcal{S}_n$ and target space $\mathcal{A}_n$.